

Shared Substrate

A Discipline for Sustained Human–AI Coupling on Complex Problems

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Abstract

Modern AI agents have characteristic limitations on long-horizon complex work: bounded context windows, drift across iterations, phantom decisions (claims of resolution not grounded in any artefact), reference rot, lost intent, and the cost of restarting cold each session. These limits will not be dissolved by larger models — they are limits of *substrate*, the externalised state any sustained reasoning must traverse. We articulate **Shared Substrate**: a discipline the author has practised across many domains and many years of sustained complex work, made explicit here in response to the demands of multi-session AI collaboration — the discipline of forming, maintaining, and cascading hierarchical external representations across the manifestation trajectory of a project. The organising image is the *centroid*: the participants in sustained work — the human operator, AI agents, successive model generations, colleagues — are noisy point masses in motion, and the substrate is the deliberately *pinned* centre of mass of that ensemble. Session drift is excursion; warm-up is reversion; the discipline is what holds the centroid stationary. The discipline rests on long-standing cognitive-science foundations — distributed cognition (Hutchins, 1995), external representations (Scaife and Rogers, 1996), cognitive load (Sweller, 1988), hierarchical levels of analysis (Marr, 1982), boundary objects (Star and Griesemer, 1989) — recovers software-engineering precedent (Architectural Decision Records (Nygard, 2011), Single Source of Truth, hexagonal architecture (Cockburn, 2005), append-only logs (Candea and Fox, 2003)) alongside emerging agentic-AI patterns (Lewis et al., 2020; Packer et al., 2024; Shinn et al., 2023; Bai et al., 2022), and is grounded formally: each abstraction layer of a project is paired with a representation formalism and a verification oracle — from executable behavioural contracts at the topmost layer to measurable non-functional envelopes at the operational one — with the work’s state kept observable to the agents themselves, closing the feedback loop that enforces guardrails while divergence is still cheap to correct. The substrate that results is best understood as a compressed representation of the thing being manifested — agentic pipelines expand it into the work, and the human deposits only what no pipeline can derive — so the discipline’s purpose has a one-line statement: separate genuine human creative input from cognitive load. The discipline defines six artefact categories — Knowledge Bases, Inventories, Data Dictionaries, Decision Records, Open Questions, Glossary — a status lifecycle, an edit protocol with explicit anti-patterns and a trivial-fix lane, and forward/reverse propagation rules across the hierarchy. Its posture is *human-governed, agent-executed*: a five-layer adoption stack progressively delegates substrate execution to agents while intent, decision, and scope authority remain with the person. The practice is domain-agnostic — software engineering appears here only as the worked instance — and practicable today with no special tooling. We claim the substrate matters as much as the model, and that this is the form of work AI raises in importance — not invents.

1 Why Substrate Matters as Much as Model

Most discussions of AI capability concern the model: its parameters, its benchmarks, the length of its context window. Almost none concern the *substrate* — the externalised state any sustained reasoning, whether organic or artificial, must traverse. This paper is about the substrate.

The thesis is simple to state and difficult to live up to. **For complex tasks that span many**

sessions, many decisions, and many abstraction layers, the substrate matters at least as much as the model. A larger model with no substrate degrades along predictable axes; a moderate model with a well-tended substrate can sustain coherence over arbitrary horizons. Bigger context windows are a quantitative response to a qualitative problem. They postpone the failure modes; they do not dissolve them.

By *substrate* we mean the persistent, externally visible state of the work — the artefacts read

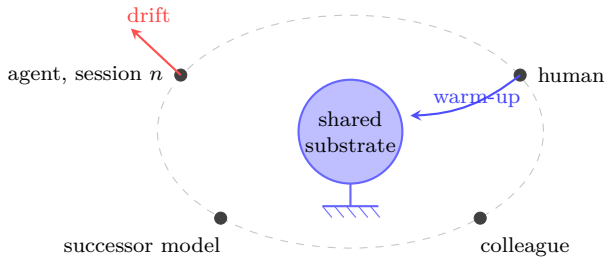


Figure 1: The pinned centroid. Participants in sustained work — human, agents, successive models, colleagues — are noisy point masses. The substrate is the deliberately pinned centre of mass of the ensemble: drift is excursion away from it, warm-up is reversion to it, and a model swap moves a point mass without moving the centroid.

at the start of a session, written to during it, and left behind for the next. Informally, the substrate is whatever remnants survive between conversations: a chat transcript, scattered notes, the source code itself. Disciplined, it is something else — a structured, hierarchical, multi-artefact graph that encodes the *current state* of the project’s understanding, with explicit semantics for each kind of artefact and explicit rules for keeping the graph coherent. The remainder of this paper concerns that disciplined practice.

We call the discipline **Shared Substrate**, and the image that organises it is the *centroid* (Figure 1). Picture the participants in a sustained project as point masses in motion: the human operator; an AI agent whose memory ends with its context window; perhaps several such agents; next quarter, a successor model; next year, a colleague joining the work. Each mass is noisy — each session’s understanding is a perturbed reconstruction of the last — and an ensemble of noisy masses left uncoupled has no fixed point: its centre of mass wanders wherever the excursions happen to carry it. That wandering has a name; it is the drift of Figure 2. The substrate is the deliberately *pinned* centroid of the ensemble — the shared centre of mass of the coupled human–AI system, held stationary by discipline rather than by any participant’s memory. Every session begins with reversion to the centroid (warm-up), works at some excursion from it, and ends by depositing its results back into it (handoff). Drift is excursion without reversion. A model swap moves a point mass; it does not move the centroid. The discipline of this paper is, in one sentence, the set of rules that keeps the centroid pinned.

An older metaphor survives in the discipline’s apparatus. The artefact graph that realises the

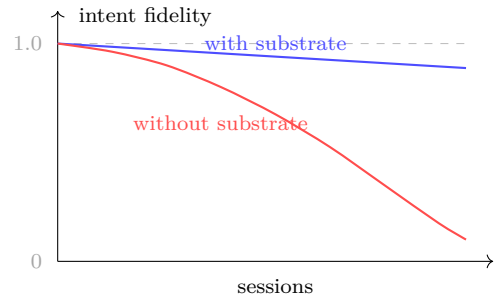


Figure 2: The drift spectrum. Without substrate discipline, intent fidelity decays superlinearly across sessions; with substrate, decay is bounded.

substrate is an *atlas* — a set of maps at multiple scales, updated by successive surveyors without losing fidelity, using conventions stable enough that someone returning years later can still read them (Figure 4). Maps shadow the territory at lower fidelity than full understanding but with substantially higher *stability* — and stability is what sustained work requires, and what the unaided memory of any single mind, organic or artificial, eventually fails to provide (Figure 2).

One more image, developed in §5, completes the set. Because the substrate is layered by abstraction — the shortest description at the top, progressively fuller detail beneath — it is a *compressed representation* of the thing being made: the small, dense, slow-changing description from which agentic pipelines expand the full manifestation. What must actually be written into it is only what no pipeline can re-derive: the choices, the intent, the taste. That is the discipline’s purpose in one line — **separate genuine human creative input from cognitive load**. The creative input is deposited once, at the layer where it belongs; the load — the re-explaining, the re-deriving, the bookkeeping — is what the substrate and its agents exist to absorb.

Two longer traditions deserve immediate credit. The cognitive-science one has known for decades that cognition is not contained in heads — that it is *distributed* across people, tools, and the environment. Hutchins’ analysis of ship navigation (Hutchins, 1995) showed that the cognitive work of plotting a course is performed by a system of people and instruments, not by any one mind. Scaife and Rogers (1996) later distinguished between *internal* representations (in the head) and *external* representations (in the environment), showing how the latter can radically reduce the cognitive load of complex tasks. The software-engineering tradition has, for at least fifteen years, exter-

nalised architectural decisions into append-only log entries (Nygard, 2011) precisely to avoid the failure modes catalogued in §2. Shared Substrate draws from both. What it adds is the systematic application to a case the older traditions could not have anticipated: where one of the collaborators is a large language model whose every session begins, in cognitive terms, with amnesia.

What follows is shaped by that observation. AI’s most discussed limitations on long-horizon work — drift, phantom decisions, restart cost — are best understood as substrate failures rather than model failures, and respond to substrate-level interventions far more reliably than to model-level scaling. The discipline that addresses them is not new; it is recovered, made explicit, and applied. And the discipline is fundamentally human in its *governance*: intent, decisions of consequence, scope, and voice remain with the person who cultivates it. Its *execution*, by contrast, is progressively delegable to the agents themselves (§6) — the same tools whose limitations motivate the practice can be enlisted to maintain the substrate that compensates for those limitations. AI’s limitations motivate the practice. Cognitive science legitimises it. Human-governed, agent-executed practice carries it out. This matters, because it means the discipline survives across model generations: the governance schema is durable while the execution layer evolves with the tooling.

The structure of what follows is straightforward. §2 catalogues the failure modes the discipline addresses and situates them against a genuinely contested empirical record. §3 traces the foundations from which it draws. §4 specifies the discipline itself. §5 pairs each abstraction layer with a representation formalism and a verification oracle, and argues that observability of the work’s state is what closes the agent’s feedback loop. §6 specifies the human-governance / agent-execution division of labour and its adoption stack. §7 maps each element of the discipline to the cognitive principle that grounds it, the failure mode it mitigates, and the layer that automates it. §8 shows the discipline in flight on two worked examples — one greenfield, one brownfield. §9–§10 discuss cost and the honest limits of the practice. §11 closes.

Provenance. The discipline articulated here was not invented for this paper, nor co-developed with an AI agent in the writing of it. It is a long-running reflection of the author across many domains and many years — a habit of

thought and work shaped by experience well outside the AI context, applied to research, to engineering, and to sustained intellectual work of several kinds. Books, lab notebooks, scientific notation, version control, architectural decision records: each generation finds apparatus for stabilising thought across time and distributing it across minds, and the author has cultivated such apparatus, in one form or another, on every complex project worth sustaining. What working with current-generation language models on complex, multi-session projects did was sharpen the apparatus into something articulable: every implicit affordance of human-only work became an explicit artefact when the collaborator was an agent without persistent memory, and the elements of the discipline became visible in a way they had not been before. This paper names what was already there. The discipline applies wherever sustained complex work crosses many sessions, decisions, or abstraction layers — not only where one of the collaborators is an AI; collaboration with AI is what most recently made the discipline’s elements demand explicit articulation, and gave us the occasion to name them. Acknowledgements at the end credit the AI’s specific role in helping draft and stress-test that articulation.

2 What We Observe When Substrate is Absent

The motivating phenomena are not theoretical. They are what one comes to see when collaborating with a large language model on any sufficiently complex task without substrate discipline. The names below are ours; the phenomena are widely reported across the practitioner community and increasingly studied formally (Liu et al., 2024; Park et al., 2023; Packer et al., 2024). Each is described from the inside — what one sees, what one feels, what fails.

2.1 Context-window saturation

Every session is bounded by a context window. Even with windows in the hundreds of thousands or millions of tokens, the working set of a long project — accumulated decisions, terminology, partial designs, prior versions — outgrows the window faster than capacity grows. The symptom is a creeping reluctance to introduce new context; a sense of crowding; an instinct to compress prior content into ever-shorter summaries.

Each compression discards material the next sub-task might have needed. Capacity addresses the *amount* of state attended to at once, not the *amount* the project has accumulated. These are different quantities, and only the second grows superlinearly with project complexity.

2.2 Lost-in-the-middle and attention dilution

When relevant content does fit in the window, it is not equally accessible across positions. The literature on long-context retrieval consistently finds that information placed in the middle of a long context is retrieved less reliably than information at the start or end (Liu et al., 2024). Felt from the human side, this is inconsistent recall: a fact stated three messages ago is honoured; a fact stated thirty messages ago is forgotten or contradicted, even though it remains technically present. Compounding this, every additional irrelevant token dilutes attention that could have been spent on the relevant ones. The agent is the same; the substrate is now noisier.

2.3 Drift across iterations

Without an external reference for the project’s settled state, each iteration is built atop the last iteration’s reconstruction of it. Small reconstruction errors compound. After enough iterations, the running understanding has drifted measurably from any historical anchor. The phenomenon shows up as a sense of “we already decided that, didn’t we?” with the answer increasingly uncertain; or as a drift in terminology, where a term used precisely in week one is used loosely by week three; or as the gentle reinvention of solutions previously rejected for reasons no longer remembered.

2.4 Phantom decisions

The most consequential failure mode. Asked “did we decide X?”, the agent answers based on a probabilistic reconstruction from available context — it cannot reliably distinguish “this was decided” from “this was discussed and was a plausible thing to decide”. Without an external decision register, the agent *will confidently confabulate*, citing reasoning that never occurred and conclusions never reached. The human side often discovers this only when the phantom decision contradicts a real one; sometimes only when implementation reveals the inconsistency. The risk is most acute precisely on

the decisions that matter most — load-bearing choices made early and referred to often.

2.5 Reference rot

In informal practice, references to prior work are by file path or paraphrase: “in the requirements doc, the section about latency”. Reorganise the file or refactor the section, and every such reference becomes stale. The agent, asked to “see the latency section”, produces a plausible reconstruction of what such a section *would have said*. The human side cannot easily distinguish reconstructed content from quoted content. References that were meaningful in week one are increasingly noise by week three.

2.6 Restart cost

Every new session begins with whatever substrate can be mustered from memory and immediately-available files. Without a stable warm-up substrate, the early minutes of every session go into re-explaining context: who is doing what, what is being built, what was decided last week, what the open questions are. The cost is borne on one side as fatigue and on the other as crowded context. In aggregate, a project that requires twenty sessions pays the restart cost twenty times.

2.7 Decided versus considered confusion

In a chat transcript there is no intrinsic distinction between “this was decided” and “this was considered”. Both look like prose. The human side relies on memory or implicit context to distinguish them. The agent, lacking that mental annotation, treats both as equivalent input. Over time, *considered* options accumulate in the running context with the same status as *decided* ones, and the agent begins to act as if they were settled. Contraries emerge silently, often resolved by the next iteration’s reconstruction in whichever direction the immediate context happens to favour.

A common shape

These seven failure modes share a single shape. None is a property of any particular model; each is a property of *unsubstantiated collaboration*. Bigger models with longer context windows shift the magnitude of each — the threshold at which it appears, the rate at which it compounds — without changing the underlying topology. Each admits

a substrate-level intervention more reliable than any model-level fix. The remainder of the paper specifies those interventions and grounds them in the cognitive-science and software-engineering traditions that have, in different idioms, addressed precisely these failure modes for decades.

The contested empirical record

The failure modes above are felt from the inside; their aggregate cost is now visible from the outside, and the picture is genuinely contested. The most methodologically careful randomised controlled trial to date (Becker et al., 2025) — sixteen experienced open-source developers, 246 real tasks on mature repositories they knew well — found that access to early-2025 AI tooling *increased* task completion time by 19%, while the same developers had forecast a 24% speed-up and, even after the experiment, still believed the tools had made them about 20% faster. Field experiments across 4,867 developers (Cui et al., 2026) found the opposite sign: a 26% increase in completed tasks, concentrated among less experienced developers. A controlled greenfield task (Peng et al., 2023) found a 56% speed-up. These results are not mutually inconsistent; they are measurements of differently configured couplings. Farrag (2026) reconciles them through three moderators: the abstraction level of the task, the maturity of the codebase, and the experience of the operator.

We read this record as the strongest available motivation for substrate discipline, for two reasons. First, the calibration failure in Becker et al. (2025) — practitioners who are slowed down while confidently believing they are sped up — means felt productivity cannot be trusted as the measure of a human–AI coupling; only externalised, inspectable state can be (§5 proposes the replacement measurand). Second, the moderators that determine which regime a coupling lands in are precisely substrate-adjacent variables: what the project’s accumulated context is, how it is represented, and at what abstraction level the human operates. The discipline of this paper is an intervention on those moderators, and that claim is falsifiable: if practised substrate discipline does not move experienced operators on mature projects out of the slowdown regime, the thesis of this paper fails.

3 What We Inherit: Foundations

The discipline did not emerge from a vacuum. Three traditions — none of which uses the term *Shared Substrate* — converge on its elements. The cognitive-science tradition supplies the deepest framing; the software-engineering tradition supplies hard-won precedent; the agentic-AI literature supplies the contemporary motivation and a small but growing apparatus.

3.1 Cognitive science

What Shared Substrate asks of a practitioner is not foreign to humans — it is what humans do when they think well at scale, made explicit and applied to a new collaborator.

Hutchins (1995) argued in *Cognition in the Wild* that cognitive work is not contained in heads. The plot of a ship’s position is computed by a system: navigators, charts, plotting tables, instrument readings, written-down conventions. No one mind holds the whole computation. The chart is not a record of the navigator’s thinking — it is part of the thinking. Shared Substrate is a particular distribution: the artefact graph is part of the project’s cognition, not a passive record of it.

Scaife and Rogers (1996) sharpened the distinction between internal representations (mental models held in working memory) and external representations (graphics, tables, diagrams, documents). External representations radically alter what cognitive operations are possible: certain reasoning steps are easy on a diagram and hard in the head. The complexity of a project’s accumulated state is not a quantity to be carried internally; it is a structure to be externalised. Risko and Gilbert (2016) corroborate this empirically: people routinely offload to calendars, lists, sketches, GPS devices when cognitive demands rise, and the offloading demonstrably frees working-memory resources for harder operations.

The need rests on hard capacity limits. Baddeley (1992) established the architecture of working memory; Cowan (2001) placed the realistic capacity at roughly four chunks, sharpening Miller (1956)’s earlier estimate. Whatever the exact number, the order of magnitude is small. A project with eighty open considerations is not held in any one mind. It must live in the substrate.

Bartlett (1932) showed that memory is re-constructive: recall is shaped by pre-organised

schemata that guide pattern-completion. An LLM asked “did we decide X?” performs an analogous reconstructive process — and like human memory, it can confabulate plausibly when the schema fits. The remedy is the same remedy human cognition has used for millennia: produce a stable external trace that can be consulted instead of reconstructed.

Marr (1982) formalised what software engineers and cognitive scientists have repeatedly rediscovered: complex information-processing problems decompose into distinct levels of analysis — computational, algorithmic, implementational. Each level has its own questions and its own appropriate language. Shared Substrate’s hierarchy of abstraction layers (Figure 3) is a Marr-shaped commitment. The brain itself, in current accounts (Friston, 2010; Clark, 2013), is a hierarchical predictive system; sustained inference is hierarchical because the flat alternative is intractable.

Star and Griesemer (1989) introduced *boundary objects*: artefacts that mean different things to different communities but enable coordination because they are robust enough to survive the variation. The artefacts of Shared Substrate are boundary objects between human and AI agent: each reads them with different prior contexts, but both can refer to the same artefact and update it without confusion. The stable identifier — KB-N, ADR-N, OQ-N — is the canonical boundary object.

Sweller (1988) formalised cognitive load theory: mental resources are spent on *intrinsic* (what the material demands), *extraneous* (what bad presentation imposes), and *germane* (what supports schema construction) load. Substrate discipline imposes intrinsic load (the discipline must be held in mind) but substantially reduces extraneous load (no more re-explaining what was decided) and increases germane load (the substrate makes patterns visible). Above a threshold of project complexity the trade is favourable; §9 addresses where that threshold lies.

3.2 Software engineering

The failure modes catalogued in §2 are not new; they have visited every team that has built complex software for more than a few months, and the tradition’s solutions have been recurrent enough to deserve names.

Nygaard (2011)’s Architectural Decision Records made the single most consequential of those solutions explicit. An ADR is a short append-only

document recording one architectural decision, its context, its rationale, the alternatives considered, and its consequences. The crucial feature is the discipline of appending: an ADR is never edited; reversal happens via a new ADR that supersedes the old. The reasoning trail is preserved, not just the current answer. Fifteen years of practice across many organisations have shown the pattern resists rot in ways ad-hoc design notes do not. Shared Substrate adopts ADRs unmodified.

Cockburn (2005)’s hexagonal architecture (Ports and Adapters) makes a related move at the architectural level: the domain depends only on abstract *ports*; concrete adapters implement those ports. The discipline produces clean separability: the domain can be tested without touching infrastructure, and adapters can be swapped without disturbing the domain. The analogue at the substrate level is the separation between artefacts of different kinds, and the discipline of cross-referencing only via stable identifiers.

Hunt and Thomas (1999) crystallised *Don’t Repeat Yourself* (DRY): every piece of knowledge must have a single, unambiguous, authoritative representation within a system. DRY is usually discussed at the level of code, but the principle generalises. In Shared Substrate, every fact has one home; everywhere else references it by id. Duplicate prose is treated as a bug — the same way duplicate code is. The duplicates will diverge, the divergence will be noticed late, and the cost of reconciling will exceed the cost of the original abstraction.

Candea and Fox (2003) proposed crash-only software: a program designed so that the only way to stop it is to crash, and the only way to start it is from a cold-start replay. The benefit is that the recovery path is the only path, and is therefore exercised on every restart rather than only in disasters. Shared Substrate adopts the same logic: every session begins by reading the artefact graph from disk, regardless of whether the previous session ended cleanly or crashed. There is no separate “warm restart” path.

These four practices — ADRs, hexagonal separation, DRY, crash-only — were developed independently for different software problems. Their convergence on a recognisably common shape — a disciplined, externalised, append-only, sparingly cross-referenced substrate that survives reorganisation — is itself evidence that the discipline is a recurrent solution. Shared Substrate names that shape and applies it where it has so far been least

practised.

3.3 Agentic AI

The agentic-AI tradition is the youngest and supplies the contemporary motivation. AI systems built to operate over long horizons — to research, plan, browse, simulate — confront the substrate problem acutely: a context window is finite, the project is not.

Lewis et al. (2020) introduced retrieval-augmented generation: rather than packing all relevant context into the prompt, retrieve the relevant pieces from an external store at query time. RAG is now the dominant pattern for grounding LLM responses in specific knowledge bases. From the substrate perspective, RAG is a partial answer: it solves “how to bring relevant context to the agent at the moment of need” but not “how to maintain that context’s coherence as the project evolves”. The substrate retrieved into the agent’s working context is only as good as the substrate sitting at rest.

Packer et al. (2024) pushed harder, treating the LLM as the kernel of an operating system with explicit memory-management primitives — paging, hierarchical buffers, persistent storage. The MemGPT analogue of working memory and long-term memory mirrors the cognitive-science distinction between internal and external representations (Scaife and Rogers, 1996). Shared Substrate’s session warm-up is a coarse version of MemGPT-style paging, performed by a human practitioner at the start of work rather than continuously by the model.

Park et al. (2023) demonstrated that long-running simulated agents required explicit memory and reflection machinery to behave coherently across simulated days. They combined a flat memory store with retrieval and reflective summarisation. The simulated agents’ need for these mechanisms is the same need a practitioner has when collaborating with an LLM over many sessions, and the same set of solutions emerges in different idioms.

Shinn et al. (2023) showed that agents which reflect verbally on their own behaviour — generating critiques and revisions before retrying — outperform agents that do not. The analogue at the substrate level is the editing protocol: the discipline mandates a pre-edit reflection (READ, IDENTIFY SSOT, IMPACT-CHECK) before changes are made. Reflection is not optional; it is the protocol.

Bai et al. (2022) proposed Constitutional AI: encoding a small set of explicit rules the agent self-applies during generation. The constitution sits outside any single response and governs all of them. Shared Substrate’s edit protocol is a constitution at the methodological level — explicit rules a practitioner self-applies whenever touching the substrate.

The convergence has since become structural. Agentic-memory systems now organise an agent’s persistent state as networks of atomic, interlinked, evolving notes — A-MEM’s Zettelkasten-style memory is explicit about the lineage (Xu et al., 2025) — and hierarchical and graph-native memory architectures recapitulate, in system design, the artefact-graph shape this paper specifies as practice. The convergence extends to the tooling vendors: by 2025–26, nearly every commercially shipping AI coding harness had independently adopted a small auto-loaded project-root instruction file as its session entry point. Practitioners cultivating a discipline, researchers designing memory systems, and vendors shipping harnesses arrived at structurally similar substrate solutions without coordinating — the same convergence argument made above for the software-engineering tradition, and the same kind of evidence that the shape is a recurrent solution rather than a stylistic preference.

The agentic-AI literature is converging from many directions on a common conclusion: long-horizon AI work requires explicit external state, retrieval and pre-fetch, reflection, and rules. Shared Substrate offers no new component over this literature; it offers a methodology that integrates them at the level the practitioner inhabits — and, in §6, a governance schema for progressively delegating substrate execution to these very systems.

4 Shared Substrate: The Discipline

The discipline of Shared Substrate is specified by six artefact categories, a set of stable cross-referencing conventions, an explicit status lifecycle, an edit protocol, propagation rules across an explicit hierarchy of abstraction layers, an anti-pattern catalogue, and a session protocol. Each piece is modest in itself; the leverage comes from their combination (Figures 3, 4).

A note on execution before the specification. What follows describes the discipline in its fully-manual baseline form — every step performed

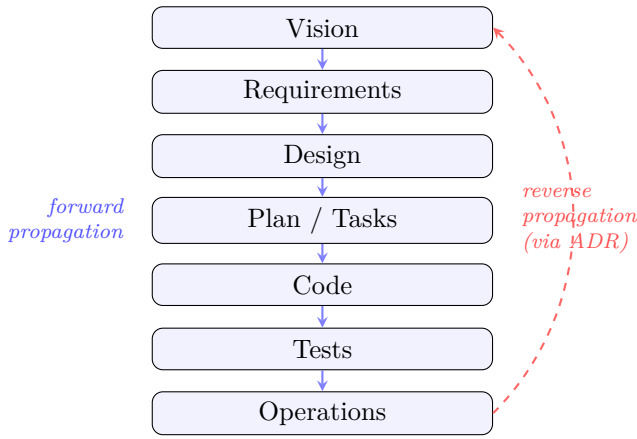


Figure 3: The hierarchy of maps. Each layer is a distinct representation; forward propagation cascades changes downward; reverse propagation requires deliberate revision via Decision Records.

by the practitioner. §6 specifies how execution is progressively delegated to agents; delegation changes *who performs* each step, never what the step is or who is accountable for it.

4.1 Six artefact categories

The substrate is composed of six kinds of artefact. Each kind has explicit semantics, an explicit format, and explicit rules about who is allowed to change it and when. The taxonomy is deliberately small: more categories would invite ambiguity about which one a given piece of content belongs to.

Knowledge Bases (KB-N) hold durable, slow-changing facts about the world the project operates in — the constraints of an external API, the regulatory environment, the assumptions of an underlying scientific tradition. KBs cite external sources with access dates and are reviewed at a cadence proportional to how fast their underlying world changes.

Inventories (INV-N) are exhaustive lists in well-defined categories: every supported order type, every emitted metric, every returnable error code. Where a KB explains, an Inventory enumerates. Inventories are designed to be machine-checkable: an automated test should be able to assert that every metric in the code is in the inventory and every entry in the inventory is in the code.

Data Dictionaries (DD-N) are schemas: every field of every domain object, with type, semantics, invariants, and example. A Data Dictionary is the artefact a tool can validate against. It is also the artefact that makes interface boundaries

explicit: two components agree on a Data Dictionary, and the dictionary becomes the contract.

Architectural Decision Records (ADR-N) record decisions, in the sense of Nygard (2011): a context, a decision, the alternatives considered, the consequences. ADRs are append-only. Reversal happens via a new ADR with a supersedes link to the old. The reasoning trail is preserved.

Open Questions (OQ-N) are the dual of decisions: deliberately unresolved questions, with target resolution dates, options under consideration, and resolution criteria. OQs prevent the silent slide from “we are deliberately not deciding this yet” into “we forgot we needed to decide this”. Once an OQ is resolved the resolution moves to an ADR; the OQ status updates to RESOLVED-BY-ADR-N; both records persist.

Glossary is the singular authority on terminology. Every term used twice in the substrate with the same meaning belongs in the glossary. If two artefacts disagree on a term’s meaning, the glossary wins or both are wrong; conversation is required. The glossary is the substrate’s semantic anchor.

The categories partition the substrate. Every piece of project knowledge belongs to exactly one of the six. The categories also have characteristic update cadences — KBs change rarely, Inventories change with each new feature, ADRs are append-only, OQs accumulate and resolve, the Glossary grows and stabilises. The differential cadence is itself useful; it tells the practitioner what kind of artefact they are looking at without reading the content.

4.2 Stable identifiers and the cross-reference graph

Every artefact carries a stable identifier of the form $\langle \text{category-letter} \rangle\text{-}\langle \text{number} \rangle$: KB-1, INV-4, DD-7, ADR-12, OQ-23. Identifiers are assigned monotonically and never reused. Cross-references throughout the substrate use these identifiers — never file paths, never section numbers, never paraphrases.

The reason is simple. File paths reorganise; section numbers shift on insertion; paraphrases drift in meaning. Each is a route to reference rot. A stable identifier survives reorganisation: KB-3 remains KB-3 whether its file is at `docs/kb/` or `kb/` or has been moved into a subdirectory. The identifier-based cross-reference is a boundary object (Star and Griesemer, 1989) between today’s substrate layout and tomorrow’s: both layouts

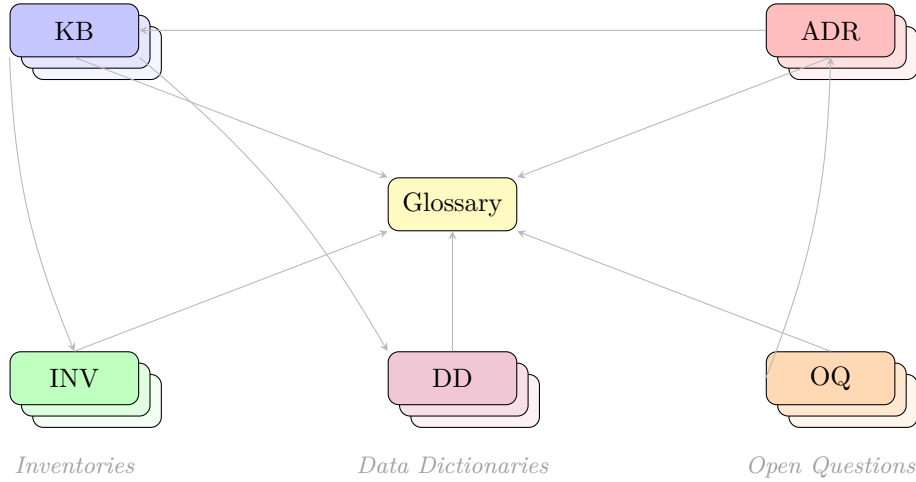


Figure 4: The atlas. An artefact graph linking Knowledge Bases, Inventories, Data Dictionaries, Decision Records, and Open Questions, with the Glossary as central semantic anchor. Stacked boxes indicate that each category typically contains many artefacts; arrows are illustrative cross-references, not exhaustive. Cross-references use stable identifiers that survive reorganisation of the file tree.

agree on what KB-3 is, even though they agree on nothing else about where it lives.

Every artefact also lists the artefacts it depends on (**depends_on**) and the artefacts that reference it (**referenced_by**). Together these turn the substrate into a navigable graph. The graph supports impact analysis: before editing KB-3, the practitioner walks KB-3.**referenced_by** and reviews each downstream artefact. The graph supports orphan detection: an artefact whose **referenced_by** stays empty for more than one milestone is a candidate for retirement. The graph supports cycle detection: two artefacts that depend on each other suggest a missing third artefact that should own the shared content.

The atlas (Figure 4) is a partial picture of the graph at a moment in time. The full graph is implicit in the **depends_on** and **referenced_by** fields of every artefact, and could be rendered automatically by tooling (§6).

4.3 Status lifecycle

Every artefact has a status, drawn from five values (Figure 5): MISSING (named in the inventory but no file yet), DRAFT (file exists, content incomplete or not yet reviewed), STABLE (content current and complete to the level required), STALE (content was once stable but the underlying world has changed or an upstream artefact has changed since last review), and DEPRECATED (superseded; kept for history, must not be referenced by new

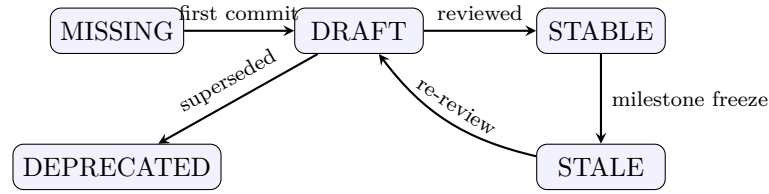


Figure 5: Status lifecycle. Transitions are explicit and triggered; no artefact silently changes state.

work).

Each transition has an explicit trigger. MISSING to DRAFT is the first content commit. DRAFT to STABLE is owner review with completeness check. STABLE to STALE is either a milestone-freeze gate (if the artefact has not been reviewed by the gate, status auto-degrades) or a propagation event from an upstream artefact that changed substantively. STALE to STABLE is re-review. Anything to DEPRECATED is supersedure.

The discipline is that the status field is mandatory and accurate. “Eventually consistent” is treated as a bug, not a feature: if two artefacts disagree, the disagreement is logged with a target reconciliation date, not silently tolerated. The same applies between an artefact’s stated status and its actual freshness — a STABLE artefact with a six-month-old review date during a fast-moving project is not stable; it is stale, and should be labelled accordingly.

The status field, combined with the cadence rules from §4.1, gives the practitioner a quick estimate of the substrate’s overall health. A sub-

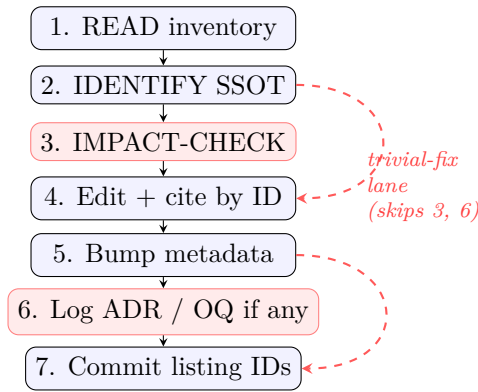


Figure 6: The edit protocol. The main path runs top-down through all seven steps. The trivial-fix lane (red dashed arrows) bypasses IMPACT-CHECK (step 3) and decision-logging (step 6), which are the steps that do not apply to typos and mechanical fixes. Steps 4, 5, and 7 always run.

strate with most artefacts in STABLE or DRAFT and few in STALE is in good shape. A substrate with many STALE or MISSING artefacts is signalling that the practitioner has out-grown the substrate or under-tended it.

4.4 The edit protocol

Every change to the substrate follows a sequence (Figure 6).

Before the edit: **READ** the inventory to confirm the artefact exists at the expected status; **IDENTIFY** which artefact owns the fact you intend to change (the Single Source of Truth, in the language of Hunt and Thomas (1999)); and **IMPACT-CHECK** by walking the artefact’s `referenced_by` list to identify downstream artefacts that may need to update.

During the edit: touch the minimum surface needed. Use stable identifiers in cross-references — never file paths, never paraphrases. Where you find yourself paraphrasing content from another artefact, replace the paraphrase with a citation. Where you find that the same fact lives in two artefacts, consolidate.

After the edit: bump `last_reviewed`; if the change is substantive, bump `version`; if status changes, update the inventory in the same commit. If the edit reflects a decision among alternatives, write the corresponding ADR in the same commit. If the edit raises a question that cannot be resolved in line, write the corresponding OQ. Commit messages list every artefact touched, by stable identifier.

The trivial-fix lane. A protocol that imposes the full sequence on every typo is a protocol that gets bypassed. A trivial-fix lane is built in: typos, formatting, broken markdown, link fixes that do not change meaning skip the IMPACT-CHECK and the decision-logging steps. The `last_reviewed` field still bumps. If you are uncertain whether a fix is trivial, it isn’t; use the full lane. The trivial lane exists to reduce extraneous cognitive load (Sweller, 1988), not to launder ceremony around non-trivial changes.

4.5 Forward and reverse propagation

Changes at one layer of the abstraction hierarchy (Figure 3) may imply work at others. The discipline distinguishes two propagation directions.

Forward propagation is the high-to-low cascade. A change to Requirements may flag downstream review of Design, Code, Tests, and Operations. The discipline mandates that forward propagation happen in the same commit if at all possible, or be filed as an explicit OQ with a target resolution date if it cannot. A lurking “I’ll update the tests later” is the road to drift.

Reverse propagation is the low-to-high direction. Implementation may reveal a constraint not anticipated in design; a test may surface a missing requirement; operations may reveal a runtime constraint that requires architectural revision. The discipline mandates that reverse propagation always pass through an ADR. A higher-layer artefact is never silently retro-fitted to match what was built; the ADR records that a lower-layer discovery has caused a higher-layer revision.

A higher-layer artefact may be *frozen* for a development phase — typically Requirements is frozen during implementation. A frozen artefact is not edited; an ADR proposing the unfreeze is required first. The protocol is heavy by design: opening Requirements during implementation is exactly the kind of action that benefits from deliberate friction.

The discipline of bidirectional propagation is what keeps the hierarchy of abstraction layers more than a label. The layers mean what they say only because changes to any layer cause review of the others, and the reviews leave traces.

4.6 Anti-patterns

Most damaging mistakes recur. Naming them and forbidding them explicitly is cheap insurance. The catalogue below is the discipline’s “do not”.

1. Editing a STABLE artefact without bumping `last_reviewed`. The change is invisible to the next reviewer.
2. Pasting the same content in two artefacts. A fact has one home.
3. Making a decision in conversation without writing the ADR. Phantom decisions are how this fails most.
4. Renaming an artefact's file without sweeping cross-references — prevented by the stable-identifier convention if the identifier remains.
5. Creating an artefact without adding a row to the inventory. The artefact becomes an orphan no one will discover.
6. Quoting another artefact's prose verbatim. The duplicate will drift; the right move is a citation.
7. Letting Open Questions accumulate without target resolution dates. The OQ register becomes a wishlist instead of a register.
8. Mixing levels — design specifics in Requirements, file paths in Design, configuration values in the Glossary. Each level has its language.
9. "I'll update the tests later." The deferred propagation becomes the next session's drift.
10. Editing the body of a past ADR. Reversal goes through a superseding ADR; the past ADR is annotated with its supersession but otherwise immutable.
11. Bypassing the SSOT rule "just for clarity". Clarity is achieved through structure, not duplication.

The catalogue is not exhaustive. New anti-patterns surface as the discipline is practised in new domains; the catalogue grows append-only.

4.7 Session protocol

The session-level discipline ensures every session begins by reading the substrate and ends by depositing state back into it (Figure 7) — in the language of §1, every session opens with reversion to the pinned centroid and closes by depositing its excursion back into it (Figure 1).

Warm-up. Before any productive work in a new session: read the inventory; read the protocol's TL;DR refresher; read the artefacts the current task touches; spawn supplementary reads (or explorer agents) for any unfamiliar dependencies. Five to ten minutes spent on warm-up substantially reduces the rate of phantom decisions, reference rot, and decided-versus-considered confusion later in the session.

Work. Apply the edit protocol (§4.4) to every

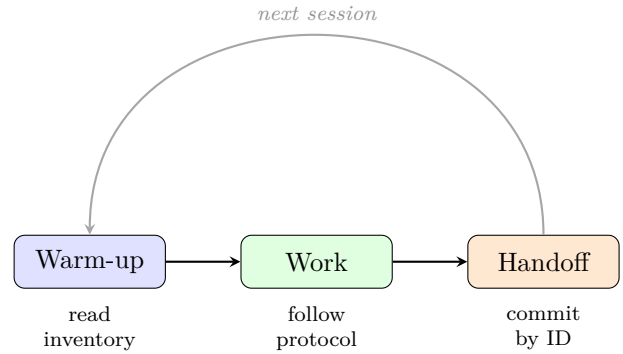


Figure 7: Session lifecycle. Every session takes the same warm-up path, whether the prior session ended cleanly or crashed. The *next session* loop is the same loop on every restart.

change. Cite artefacts by stable identifier in conversation, not by paraphrase. Use task-tracking primitives to make multi-artefact edits explicit and atomic. When in doubt about which artefact owns a fact, stop and confirm before editing.

Handoff. Before declaring the session done: every edit is committed; every new MISSING artefact created has a stub frontmatter and an inventory row; every decision has its ADR; every raised question has its OQ; `last_reviewed` and `version` are current on touched artefacts; the inventory's priority queue reflects the new state. The next session's warm-up is the previous session's handoff, read by whoever (or whatever) is doing the next session.

The protocol applies symmetrically across the human and AI collaborators. Both read the substrate at session start. Both update it during work. Both leave it consistent at session end. The asymmetry — that the AI agent's session is bounded by its context window in a way the human's is not — is precisely what the discipline is designed to compensate for.

5 Every Layer Has a Language and an Oracle

The hierarchy of Figure 3 says that layers exist and that changes propagate between them. This section says what a layer is *made of*. The claim is that every layer of the hierarchy should be paired with two things: a **representation formalism** — the language in which that layer's content is stated — and an **oracle** — the strongest available mechanism for checking work against that layer's stated intent.

The motivation is sharpest in the AI setting. Lahiri (2026) names the central bottleneck of AI-era work the *intent gap*: the semantic distance between what the operator means and what the produced artefact does. Generated output is plausible by construction, not correct by construction, and AI widens the gap by producing volume faster than scrutiny can keep up. Crucially, “there is no oracle for specification correctness other than the user”: at the topmost layer, only the human can say whether the stated intent is the intended intent. Every layer below that can and should have a mechanical or semi-mechanical oracle. The discipline’s layered substrate is how the user’s scarce oracle capacity is spent where nothing else can substitute for it, while cheaper oracles police every other layer.

The principle is domain-general. In empirical research, the top layer’s oracle is a pre-registered hypothesis tested against held-out data; in quantitative modelling, agreement with known analytical solutions; in regulatory drafting, traceability of every clause to its source obligation. We instantiate the principle for software (Table 1) because software currently has the most mature oracle stack — not because the discipline is about software.

Why a tower is sound. The pairing would be decorative if changes could not be trusted to cross layers. Abstract interpretation (Cousot and Cousot, 1977) gives the theorem-shaped justification: sound approximation between a concrete and an abstract domain is a Galois connection, and the composition of two Galois connections is itself a Galois connection — so a tower of abstraction layers preserves soundness layer-by-layer if each adjacent pair does. This reframes agent divergence precisely: an agent diverges when it violates the semantic-preservation obligation between the layer it was instructed at and the layer it produced. The discipline’s forward and reverse propagation rules (§4.5) are the operational form of maintaining those connections.

What guardrails can and cannot do. At the implementation layer, grammar-constrained decoding can provably force generated output into a target grammar (Park et al., 2025). The guarantee is syntactic only — constraint is not correctness, and it is not even safety: a benign-looking grammar constraint can itself be used to jailbreak a model into producing malicious code (Zhang

et al., 2026). Verification-aware pipelines go further — Dafny as a hidden intermediate language compiles verified code to the target language (Li et al., 2025), and LLM-generated formal postconditions have caught dozens of real historical bugs while remaining far from complete coverage (Endres et al., 2024). The stack is real but young, and it enforces nothing above the layer it operates on. This is the argument for layered oracles rather than any single mechanism: each layer’s guardrail is blind to the layers above it.

Observability closes the loop. An oracle the agent cannot query is not a guardrail; it is an audit. For guardrails to be enforced *in the loop* — during the work, not after it — the state of the work must be observable *to the agent*: measurements, traces, and check results exposed as channels the agent can consult cheaply, repeatedly, and without human mediation (Figure 8, left). In the software instantiation these are logs, telemetry, and oracle runs behind queryable service endpoints and tool interfaces; in research, analysis outputs and their provenance; in a physical design, simulation and measurement results. The form varies; the requirement — that the thing being made can be interrogated about its actual state, by the agent, at any moment — does not. The reason is the amplifier structure of the coupling. An agent is gain: it multiplies operator intent into volume. But gain is indifferent to sign — the same amplification that compounds progress compounds divergence — and an amplifier run open-loop diverges at the rate of its gain (Figure 8, right). We suggest this is the mechanism behind the slowdown regime of §2: couplings that turn net-negative not because the model is weak, but because generated volume runs unobserved long enough that correcting the divergence costs more than the generation saved. A closed loop inverts the economics: each check detects excursion while it is small, so the cost of correction stays proportional to the excursion rather than to the volume built on top of it — in centroid terms, observability is what makes reversion cheap. And the loop compounds across sessions through the substrate: what each check finds — violations, escalations, the decisions they forced — is deposited as substrate records, so the fast within-session loop feeds the slow across-session loop that the substrate carries (Figure 11). The human’s place in this loop is not sensor but *designer*: choosing the contracts and budgets of Table 1, the thresholds at which

Layer		Representation formalism	Oracle	Guardrail
Business behaviour	be-	executable behavioural contracts (Gherkin scenarios)	acceptance runs against the live system	behavioural lock-in of the incumbent before any replacement work
Requirements		structured natural language with stable-ID references	human review of a formal re-statement — the one oracle only the user can be	frozen status; ADR-gated unfreeze
Design / contracts	con-	Data Dictionaries, schemas, Decision Records	schema validation; contract tests	SSOT; propagation rules
Implementation		typed code	compiler and type checker; property tests; postconditions	syntactic constraints on generation (grammar-constrained decoding)
Non-functional envelope		measurable NFR contracts: latency, memory, throughput budgets	runtime instrumentation of the incumbent and the replacement	budget assertions in CI; canary comparison against the incumbent

Table 1: The software instantiation of layer $\rightarrow$ (representation, oracle, guardrail). Other domains substitute their own columns; the discipline supplies the pairing, not the particular formalisms.

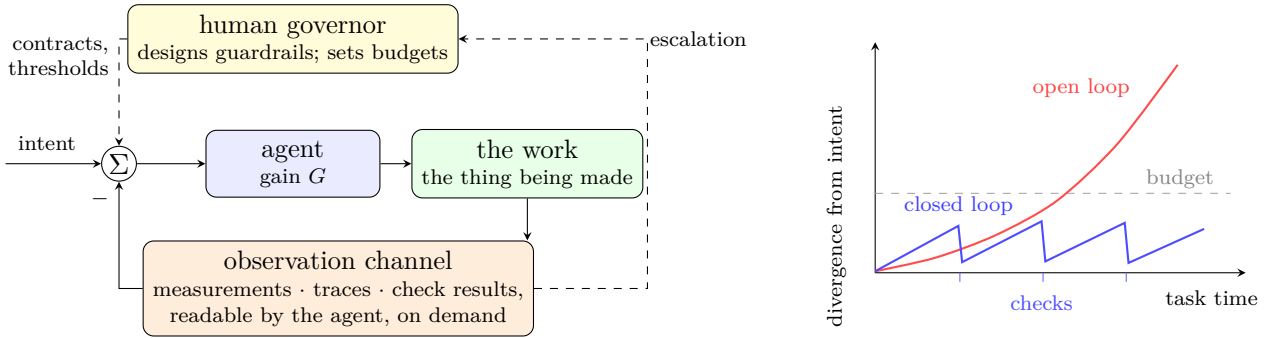


Figure 8: Observability closes the loop. *Left:* the coupling as a control loop. The human governor designs the guardrails — contracts, budgets, escalation thresholds — but does not sit in the loop as its sensor; the agent is the forward-path gain; the work’s state feeds back through an observability channel the agent can query directly. *Right:* the consequence. Amplification is indifferent to sign: run open-loop, the gain that compounds progress compounds divergence at the same rate; run closed-loop, each check detects the excursion while it is small, and correction stays cheap.

the agent must stop and escalate, and the cadence of checks. Efficient coupling is a control-design problem. Give the operator controls that cap excursion early, and the amplifier works for you; omit them, and it works against you — at the same gain.

The non-functional envelope. The formal literature on constraining AI-generated work is overwhelmingly about functional correctness; the non-functional envelope — latency, memory, throughput, operational cost — is nearly absent from it. Yet in brownfield work the envelope is often the binding constraint, and it admits an oracle most projects already own: *the incumbent system*. Before replacing a latency-sensitive process, capture its observable behaviour as executable contracts *and* its runtime profile as measured budgets —

on a JVM, allocation and memory behaviour are recoverable from JMX-level statistics, or at finer resolution by bytecode-level instrumentation of allocation sites — and hold the replacement to those budgets continuously. The incumbent becomes the oracle at every layer it can speak for. §8 shows this in flight. Evidence on LLM-generated behavioural specifications points the same way: quality ratings are high, but omissions and hallucinations occur at rates that require systematic human review (Hassani et al., 2025) — which is exactly the layered posture: the mechanical oracle runs everywhere it can, and the human oracle reviews where only the human can.

The substrate as compressed source. There is a reading of the whole apparatus in one information-theoretic sentence: the substrate is a

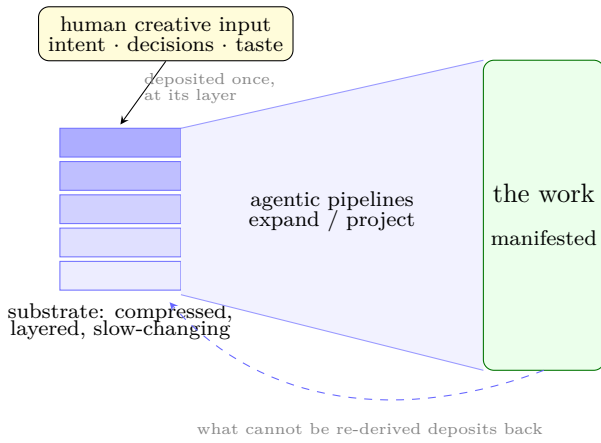


Figure 9: The substrate as compressed source. The strata are the hierarchy of Figure 3: shortest description on top, derivable detail beneath. The human deposits only what no pipeline can re-derive — intent, decisions, taste — once, at the layer where it belongs; agentic pipelines expand the compressed representation into the manifested work, with the oracles of Table 1 bounding the distortion of the expansion (Figure 8); discoveries that cannot be re-derived deposit back. Everything else is cognitive load — exactly what the human no longer carries.

compressed representation of the work being manifested, and the agentic pipeline is its decompressor (Figure 9). The hierarchy of layers makes the compression progressive: the top layer is the shortest description, and each layer beneath expands it with detail that is — crucially — *derivable*, given the layers above and the facts on record. What must actually be stored is what no pipeline can re-derive: the choices among alternatives, the intent behind them, the taste that selected this design over that one. It is no accident that the append-only Decision Records are the substrate’s most protected artefacts; they are the incompressible kernel. The reading has a standard formalisation. In algorithmic information theory, the information in a deposit x that is derivable from a substrate S is captured by conditional complexity $K(x | S)$ — the length of the shortest program producing x given S as input (Kolmogorov, 1965; Li and Vitányi, 2008). Relative to a pipeline d , a contribution is *derivable* exactly when $K_d(x | S)$ is negligible, and the creative kernel of a deposit is its conditional description length given the prior substrate and the pipeline. The pipeline-relativity is a feature, not a defect: as pipelines strengthen, more becomes derivable and the kernel shrinks — the formal shadow of the operator’s migration up the abstraction hierarchy. This gives the discipline’s purpose its sharpest statement: the entire

apparatus exists to **separate genuine human creative input from cognitive load**. The human supplies the bits that cannot come from anywhere else — once, at the layer where they belong; everything derivable — the expansion, the bookkeeping, the propagation, the re-derivation that flat collaboration forces the human to repeat every session — is carried by the substrate and executed by the pipeline, with the oracles bounding the distortion of the expansion. The failure modes of §2 read naturally in this frame: drift is decoder error accumulating unchecked; a phantom decision is the decoder inventing source bits; restart cost is retransmission of content that should have been stored compressed.

What productivity means here. The contested record of §2 is partly a dispute about the measurand. Classical productivity — output volume per unit time — was a serviceable proxy while production was the scarce step. In a coupled environment it fails twice over: volume is precisely what the amplifier inflates, and it inflates divergence as willingly as progress; and perceived productivity tracks volume, which is why practitioners in the measured slowdown regime sincerely felt faster. The frames of this section supply the replacement. The scarce inputs of a coupling are three: the human’s creative kernel (the non-derivable bits), the human’s oracle capacity (review and approval where no mechanical oracle can substitute), and the calendar time of the loop. The genuine output is not generated volume but *validated intent made durable*: expansions that survived their layer’s oracle, and the residue of the session — decisions recorded, questions resolved, maps updated — deposited into the substrate so that it is never paid for twice. Productivity, in the environment this paper proposes, is the rate at which durable, validated intent accumulates per unit of scarce human input. Generated volume drops out of the measure entirely; on the trajectory scale, unverified volume is not output at all — it is rework not yet scheduled.

From definition to measurement. K is uncomputable, but the definition admits computable surrogates — and the discipline is self-instrumenting: every term of the measure corresponds to an event stream the substrate already externalises. Kernel content is estimable by minimum-description-length surrogates (Ris-

sanen, 1978) or, directly practicable today, by the code length a strong language model assigns to a deposit conditioned on the prior substrate, $-\log p(x \mid S)$ — language models are, formally, general-purpose compressors (Delétang et al., 2023) — so derivable content scores near zero while a direction-changing decision scores high. Validation weight is the locked contracts’ *discriminating power*: the fraction of injected behavioural mutations they catch (Endres et al., 2024), which resists inflation by padding in a way raw contract counts do not. Durability is survival: the fraction of deposited artefacts still referenced and un-invalidated k sessions later, together with the warm-up cost curve, which stays flat under working durability and grows with project size without it. The denominator is logged by construction — kernel time, oracle time (approvals at L2 are events), load time. Unverified volume outstanding is the liability account, and churn attributable to failed oracles is the debt coming due. None of this needs apparatus beyond §6: the metering layer is the L1 integrity watchdog, read for a second purpose.

A research horizon, flagged as such. The pairing in Table 1 is stated, not derived, and the estimators above are surrogates, not theory. A formal treatment would model the human-agent channel in rate-distortion terms — rate as specification effort, distortion as residual intent misalignment — for which program-induction work already supplies a skeleton (Zhou et al., 2024), with Felleisen (1991)’s expressiveness giving the complementary inter-layer measure. On that view, the optimal granularity of a layer’s representation is not fixed: it moves with the ability of the team operating the coupling. We flag this deliberately and do not develop it here; the discipline stands on the weaker, practicable claim that *some* explicit pairing at every layer beats an implicit one at any.

6 Human Governance, Agent Execution

The discipline’s *content* — six artefact categories, stable identifiers, the status lifecycle, the edit protocol, propagation rules, anti-patterns — is fixed by §4. Its *execution model* is not. As specified, every step is performed by the practitioner; left there, the discipline would carry a manual burden that does not scale with project complexity,

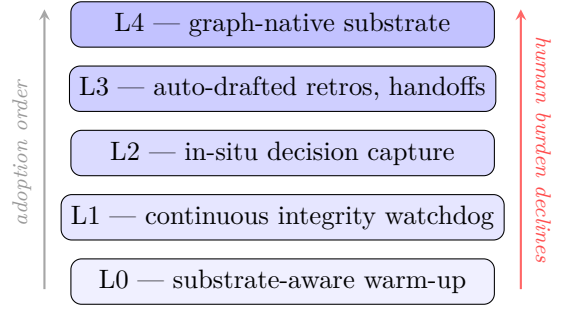


Figure 10: The adoption stack. Each layer delegates more substrate execution to agents; the discipline’s semantics are unchanged at every layer. L0 and L1 adopt first; L4 is the discipline’s mature form. The human’s residual surface — intent, decision approval, scope, voice — does not delegate.

and would invite the objection that autonomous memory systems will make it obsolete. The convergence documented in §3.3 argues the opposite posture: Shared Substrate is the **human-governance schema** over substrate execution, agnostic to who — or what — performs the execution.

The delegation is codified as a five-layer adoption stack (Figure 10).

L0 — substrate-aware warm-up. The session warm-up of §4.7, performed by the agent instead of the human. The static baseline costs nothing: modern agent harnesses auto-load a small project-root instruction file at session start, and that file — versioned, edit-protocol-governed, a *pointer* to the canonical artefacts and never a restatement of them — turns the harness’s existing behaviour into the warm-up entry point. The pattern is agent-agnostic in semantics and platform-specific only in filename; the discipline commits to the pattern, not to any vendor’s loader. The dynamic form is richer: an agent that walks the `depends_on` closure of the current task and reads what it finds.

L1 — continuous integrity watchdog. Background checks that need no judgement: every cross-reference resolves to an artefact whose frontmatter `id` matches; every file has an inventory row and every row a file; frontmatter schemas validate; STABLE artefacts unreviewed at a milestone freeze auto-degrade to STALE; orphans and cycles are flagged. None of this is novel tooling — linters, schema validators, graph checks — and none of it requires judgement, which is exactly why it delegates first. L1 is the substrate’s own observability channel: the closed loop of §5 pointed at the artefact graph itself.

L2 — in-situ decision capture. The agent

drafts the ADR at the moment a decision is made in conversation, rather than relying on the human to remember to write it afterwards. The human approves before the draft becomes a record: an unapproved auto-drafted ADR is not a decision, it is a phantom decision with better formatting. L2 targets the single most consequential failure mode (§2.4) at its source.

L3 — auto-drafted retrospectives and handoffs. Session summaries, next-session prompts, and the inventory’s priority queue populated from observable state — commits, artefact touches, resolved and raised questions — with the human editing rather than authoring.

L4 — graph-native substrate. The artefact graph held as a first-class knowledge graph, with the markdown artefacts as rendered views and the atlas of Figure 4 drawn by tooling rather than by hand.

Adoption order matters: L0 and L1 first — low cost, immediate utility, independent of each other; L2 and L3 only after L0/L1 have proven reliable, because they draft content in the human’s voice; L4 is the discipline’s mature form, not its entry point. At every layer the human’s residual surface is the same: **intent declaration, decision approval, scope governance, and voice authority over substantive prose.** That surface does not delegate — it is what governance means. The stack changes who executes the discipline; accountability for the substrate, like authorship of the project, stays with the person. Seen through the compression reading of §5, the stack is a distillation apparatus: layer by layer it strips cognitive load away from the person until what remains is only the input no pipeline can generate — the creative kernel. Delegation is not the human doing less; it is the human’s effort converging on the only work that was ever irreducibly theirs.

Figure 11 assembles the whole coupled system in one picture, deliberately free of any domain’s vocabulary. Read it as the paper in miniature. The work being manifested — a paper, a design, a device, a strategy, a codebase — advances under two nested loops. The fast loop is §5: within a session, agents act on the work and observe its actual state against the layer contracts. The slow loop is §4.7: across sessions, every participant reverts to the substrate at warm-up and deposits back at handoff — and what the fast loop finds, the slow loop keeps, because violations, escalations, and the decisions they force are themselves deposited as substrate records. The human sits

above both loops, governing; the substrate sits beneath both, evolving. It is the only element in the picture that neither resets nor retires: agents are swapped, sessions end, the work ships — the evolving external memory carries the coupling’s accumulated understanding forward into whatever comes next.

7 The Synthesis: Discipline ↔ Cognitive Principle ↔ Failure Mode

This section is the paper’s central synthesis. Each element of Shared Substrate from §4 maps to a cognitive principle from §3 that grounds it, to a failure mode from §2 that it mitigates, and to the adoption-stack layer (§6) that can execute it on the practitioner’s behalf. Table 2 lays out the mapping.

The shape of the table matters as much as its content. Every discipline element has a cognitive precedent, and every discipline element addresses an observable failure. The discipline is not arbitrary, and it is not a list of best practices. It is a small set of structural choices, each with two sources of justification: why it should work (the cognitive principle) and what happens when it is absent (the failure mode).

Three readings of the table are useful. From left to right, it answers “why these elements?”. From right to left, it answers “what to use against this failure?”. Down the final column, it answers “what must I still do myself?” — and the answer is short: govern the hierarchy, approve the decisions. The redundancy is the synthesis’s robustness: every element is on the table for independent reasons.

A reader pressed for time should read §1 and Table 2. The argument of the paper survives the reduction.

8 Two Worked Examples

We illustrate Shared Substrate twice: once on a greenfield multi-month research project, once on a brownfield enterprise replacement under a hard time-box. In both cases specifics are generalised, and in both the engineering content is not the point. The point is the substrate trajectory.

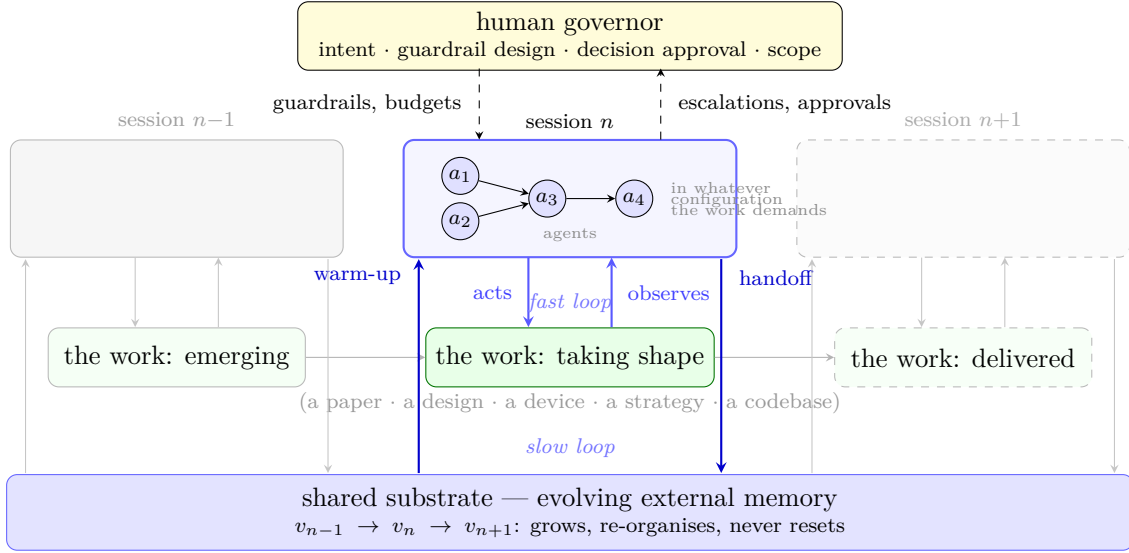


Figure 11: The coupled system in action, in no domain’s vocabulary. The human governs; agents — in whatever configuration the work demands — act within a session; the thing being made advances left to right. Two nested feedback loops drive it: the *fast loop* within a session (agents act on the work and observe its actual state against the guardrails, §5), and the *slow loop* across sessions (warm-up reverts to the substrate, handoff deposits back, §4.7). Sessions end and agents are swapped; the evolving substrate never resets — it carries the coupling’s accumulated understanding forward.

8.1 Greenfield: a research-to-production bridge

The first project was a live algorithmic-execution engine bridging a research backtesting framework to a regulated broker.

The project began as a chat. A practitioner described the problem: a backtesting framework that produces strategies, a broker API to send them to live, the gap between the two. The first artefact deposited to the substrate was a Requirements document — flat, around six thousand words, the practitioner’s first pass at what would be built and why.

The second artefact was a Context Inventory, listing the categories of substrate the project would need: Knowledge Bases, Inventories, Data Dictionaries, Decision Records, Open Questions, Glossary. The inventory was at first largely MISSING; it named the artefacts that would be needed without yet writing any.

The third artefact was a Context Protocol — the discipline document specifying how the substrate would be edited, how cross-references would work, what the status lifecycle was. The protocol came before most of the content because it determined how the content would be added.

From there the substrate grew. The first Knowledge Base captured the strategy taxonomy — the archetypes the project would support. That KB raised ten Open Questions about scope. The

practitioner made decisions on those OQs in a single working session; the decisions became the project’s first set of Architectural Decision Records. The KB updated to mark the OQs as RESOLVED-BY-ADR; the OQs were not deleted, just re-stated.

Knowledge Bases on the broker’s API capabilities and pacing limits followed. A Cost Dictionary on the project’s cost models. A Companion Projects map clarifying boundaries with related work. A Frameworks Survey reviewing existing solutions and recording which were adopted, studied, or rejected — and why. Within a few working sessions, thirteen Knowledge Bases existed; three Inventories; nineteen Architectural Decision Records; twenty-two Open Questions; one Glossary that had absorbed the terminology sections of multiple other artefacts.

The Requirements document, originally a flat draft, underwent a v0.2 pass: sections that had been narrative summaries of survey content, decision rationale, and open questions were collapsed into stable-identifier references to the relevant Knowledge Bases, Architectural Decision Records, and Open Questions. The document became substantially shorter, more authoritative, and easier to update — because the content it had been carrying now lived in artefacts whose discipline was to be authoritative.

The substrate trajectory is the point. The project did not begin with a plan and execute

Discipline element	Cognitive principle	Failure mode addressed	Executed by
Hierarchy of abstraction layers	Marr’s three levels of analysis (Marr, 1982); hierarchical predictive coding (Clark, 2013)	Drift across iterations; lost intent	human; L4
Single Source of Truth + DRY-for-prose	External representations (Scaife and Rogers, 1996); cognitive load (Sweller, 1988)	Decided-vs-considered confusion; reference rot	L1
Stable identifiers (KB-N, ADR-N, OQ-N)	Boundary objects (Star and Griesemer, 1989); persistent symbolic anchors	Reference rot; phantom decisions	L1
Append-only Decision Records	Distributed cognition (Hutchins, 1995); preservation of reasoning trail	Phantom decisions; silent reversal	L2 + human approval
Status lifecycle (MISSING → DRAFT → STABLE → STALE)	Schema theory (Bartlett, 1932); affordances of state	Drift; stale knowledge	L1
Edit protocol with trivial-fix lane	Cognitive load theory (Sweller, 1988); chunking (Miller, 1956; Cowan, 2001)	Restart cost; over-ceremony abandonment	L1–L3
Glossary as authoritative semantic anchor	Encoding specificity (Tulving and Thomson, 1973)	Terminology drift	L1
Session warm-up + handoff protocol	Cognitive offloading (Risko and Gilbert, 2016)	Restart cost; context-window saturation	L0, L3

Table 2: The synthesis. Each discipline element, the cognitive principle that grounds it, the failure mode (§2) it addresses, and the adoption-stack layer (§6) that can execute it. “Human” marks work that does not delegate.

it. It began with the cheapest possible substrate (a Requirements document) and grew the substrate as the work demanded it, with the discipline ensuring that every growth step left the artefact graph coherent. By the time implementation began, the substrate was sufficient to absorb a session’s worth of work without drift, and the practitioner could resume after a break of weeks without reconstruction cost.

The metrics — thirteen KBs, nineteen ADRs, twenty-two OQs — are incidental. The trajectory — substrate as living artefact, growing to match the project’s demand for stability — is the substance.

8.2 Brownfield: replacing a legacy process in a regulated enterprise

The second example is brownfield, and adversarial in the way enterprise work usually is. At a regulated financial institution, the author replaced a legacy low-latency Java process — a component that had accreted responsibilities for years, did too much, and had become hard to evolve — with a new implementation stood up from scratch in roughly two weeks, the second of which was spent almost entirely inside the non-functional envelope: allocation behaviour, memory footprint, latency profile.

The trajectory, not the speed, is the point, and it followed the shape of this paper. The work began not with code but with a top-down design investigation: recovering what the incumbent actually did — as opposed to what its documentation said it did — and an explicit phased plan for standing the replacement up and retiring the incumbent. The incumbent’s observable behaviour was locked in as executable contracts before replacement work began, so that “done” meant *equivalent where equivalence was intended* rather than *looks right*. The non-functional envelope was locked the same way (§5): the incumbent’s memory and latency profile captured by runtime instrumentation — JMX-level statistics first, bytecode-level allocation capture where finer resolution was needed — and held as budgets the replacement had to meet continuously, queryable in-loop so that a violation surfaced during implementation rather than at review (Figure 8). Inside those guardrails, agent-assisted implementation was fast precisely because it was safe: the incumbent was the oracle at every layer it could speak for, and each phase of the plan mapped to one abstraction layer with its own representation and its own oracle.

What the time-box demonstrates is where the

binding constraint was. Code generation was never the bottleneck — the agent produced implementation quickly. The rate-limiting step was locking intent, layer by layer, fast enough to make the generated volume safe to accept. That inversion — specification and verification as the critical path, generation as the cheap step — is the practical content of §5, observed in the field.

Neither example proves the discipline works; each shows what it looks like in flight. Both are single-practitioner accounts, reported for their shape rather than their effect sizes — a published case study of an AI-augmented one-person squad rebuilding inside a brownfield enterprise (Vilas Boas et al., 2026) suggests the shape generalises, but the controlled evidence of §2 is the standard the claim must eventually meet. And neither example separates the contribution of the discipline from the contribution of the practitioner applying it — the hidden expertise requirement named in §10.

9 The Practice in the Hand

Shared Substrate is not free. The discipline costs effort, and the effort is borne by the practitioner. This section names those costs honestly.

The protocol’s heaviest cost is at the start. The first substrate-disciplined session on a new project takes longer than the same session run as flat chat, because the inventory is MISSING and must be created, the protocol must be set up, and the practitioner is doing two things at once: working on the project, and building the substrate the project will live in. This investment pays back roughly from session three or four, when the substrate begins to absorb context that flat chat would otherwise have demanded re-establish each session.

The ongoing cost is intrinsic load (Sweller, 1988): the discipline must be held in mind. The pre-edit READ, the IDENTIFY SSOT, the IMPACT-CHECK — none of these is hard, but together they impose a small overhead on every change. The trivial-fix lane absorbs much of the overhead for small changes; the rest is amortised across the changes the discipline saves elsewhere.

There is also a hidden requirement: someone must hold the substrate in mind even while distributing it. The practitioner who maintains a substrate at the level of discipline this paper specifies has spent years building that capability across many projects, in domains both AI-related and

not. The discipline articulated here is not Monday’s lesson; it is the externalisation of years of internal habit.

The threshold above which Shared Substrate pays back is roughly: the project will span more than a handful of sessions, or more than a handful of decisions of consequence, or more than one collaborator. Below that threshold, flat chat and unstructured notes are appropriate. Above it, every additional unit of complexity exacerbates the failure modes of §2 and therefore widens the margin by which the discipline pays.

The discipline scales down naturally. Not every project needs thirteen Knowledge Bases; some need three. Not every protocol clause is invoked on every change; the trivial-fix lane is built for the common case. The aim is not to maximise discipline; it is to match discipline to project, with the project’s complexity as the dial.

Finally, the cost picture above assumes the fully-manual baseline. The adoption stack of §6 changes its shape: L0 removes the warm-up’s friction (though not the human’s need to read), L1 removes the integrity sweeps, and L2 and L3 replace authoring decision records and handoffs with reviewing drafts of them. The costs do not fall to zero — reviewing an auto-drafted ADR is cheaper than writing one, but approval is real work, and approval fatigue is a real budget — yet the residual cost concentrates exactly where the discipline wants the human’s attention: on governance.

10 Honest Costs and Limits

A self-respecting methodology paper acknowledges where the methodology fails. Shared Substrate is not universally helpful. This section names where it doesn’t help, where it costs more than it returns, and where the assumptions break.

Truly novel research. The discipline assumes the project’s hierarchy of abstractions can be articulated. In genuinely novel research — where the right ontology is not known until the work is mostly done — the substrate cannot be designed in advance, and forcing one is harmful. The discipline can be applied retroactively, after the right structure has emerged, but it cannot accelerate the emergence. For exploratory work, flat notes are often better; substrate discipline applies once the work crosses into elaboration.

Bureaucracy risk. A discipline whose ceremony exceeds its benefit is a discipline that gets

ignored or performatively followed. The trivial-fix lane is built specifically to keep this risk manageable, but the lane only works if it is generously sized. A practitioner who narrows the trivial lane until every change requires the full sequence will find their team — or their agent collaborator — quietly bypassing the discipline. The discipline must scale down or it scales out of relevance.

Substrate-author expertise as hidden requirement. A substrate is no better than the discipline of its author. The discipline articulated in this paper looks straightforward on the page; in practice it requires judgement honed across many projects. A practitioner attempting Shared Substrate for the first time will produce a worse substrate than the same practitioner three projects later. We have no way to shortcut this: the discipline is teachable, but the judgement is practised.

Multi-author coordination. The discipline as specified here assumes a single primary substrate author, with collaborators (human and AI) operating against the substrate. Multiple authors editing the substrate concurrently introduce coordination problems — branch-merge semantics, conflicting status changes, duplicate ADRs — that this paper does not address. The discipline extends to multi-author settings, but the extension is its own work.

Agentic execution adds its own failure modes. The delegation of §6 is not free. An auto-drafted ADR can record a decision that was never made — the phantom decision reborn with better formatting — which is why approval is a gate, not a courtesy. Auto-drafted prose gradually shifts framing and voice toward the model’s priors. A trusted watchdog invites the human to stop reading what it guards, and a well-automated stack invites the governance surface to shrink by neglect. Two agents executing against the same substrate can disagree; the substrate records the conflict, but a human must resolve it. Every layer of the stack changes who executes — none changes who is accountable.

The discipline is not a thinking aid. Shared Substrate helps a practitioner who has something to say organise and sustain it. It does not generate ideas, evaluate alternatives, or verify correctness. A well-organised wrong project is still wrong. The discipline preserves the practitioner’s reasoning; it does not produce it.

Model capability still matters. We have argued that substrate matters as much as model.

We have not argued that model capability is irrelevant. A model that cannot follow instructions, cite stable identifiers, or refrain from hallucinating in spite of contradicting evidence cannot be a useful collaborator on a disciplined substrate. Substrate amplifies model capability; it does not replace it.

These caveats do not invalidate the discipline. They locate it. Shared Substrate is the right tool for sustained complex work with a capable AI collaborator, above a complexity threshold, where the project’s structure is articulable. Outside that envelope it costs more than it returns, and we do not recommend it.

11 A Cognitive Practice for the AI Era

We close on the deepest implication.

AI raises the importance of work humans have always needed to do. Externalising thought, structuring it hierarchically, preserving reasoning trails, naming things stably — none of this is new. Books, libraries, encyclopedias, scientific notation, version control, lab notebooks, architectural decision records: every generation has produced apparatus for stabilising thought across time and distributing it across minds. The apparatus has worked.

What is new is the collaborator. A large language model is the first collaborator with the cognitive capacity of a careful specialist and the persistent memory of a goldfish. The asymmetry is unique: any session begins with the agent’s full intelligence intact and the project’s accumulated context absent. Substrate is how the absence is repaired, every session, indefinitely. It is also how the human’s effort is conserved for what no collaborator can supply — the creative kernel the whole system exists to amplify.

The discipline that addresses this is not invented for it. The discipline is recovered, from cognitive science, from software-engineering, and from agentic-AI literatures, and made explicit. The practice is fundamentally human in its governance: a person cultivates it, sets its intent, approves its decisions, and answers for its scope — while its execution is progressively delegated to the agents themselves. The practice survives across model generations not because it must remain manual, but because the governance schema is durable while the execution layer evolves with the tooling. It is about how minds collaborate

on complex problems, not about which model is current.

Substrate is the leverage point currently most under-practised in AI work. Model capability per token will continue to grow, and context windows will continue to lengthen, and benchmarks will continue to shift. None of this changes the substrate problem. The complexity of any project that matters grows faster than any plausible future model can absorb in a single context. The projects that succeed will be the ones whose substrate keeps pace — whoever, or whatever, performs the keeping.

Shared Substrate is one form of the practice. There may be others. What matters is that the practice exists, that it is explicit, that it is taught, and that it is shared. We have described one form. We expect the form to evolve. The form’s evolution is itself a forward-propagation event the discipline foresees.

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A The Practitioner’s Compact Field Manual

This appendix collects the discipline in a form a practitioner can print and pin. Repetition is by design: the protocol’s steps are short enough that the appendix should be readable on its own.

Frontmatter (every artefact)

id: KB-NNN | INV-NNN | DD-NNN
| ADR-NNN | OQ-NNN
title: human-readable title

status: MISSING | DRAFT | STABLE
| STALE | DEPRECATED
owner: Name | shared
last_reviewed: YYYY-MM-DD
version: 0.1
sources: [URLs with date accessed]
depends_on: [ID, ID, ...]
referenced_by: [ID, ID, ...]

Edit protocol (every non-trivial change)

1. READ the inventory; locate the artefact.
2. READ its frontmatter.
3. IDENTIFY the SSOT for the fact you intend to change.
4. IMPACT-CHECK by walking `referenced_by`.
5. Decide the lane: trivial / normal / decision-bearing.
6. Edit; cite by stable identifier; do not restate.
7. Bump `last_reviewed`; bump `version` if substantive.
8. Update inventory if status changed.
9. Log decision as ADR if applicable.
10. Log question as OQ if applicable.
11. Commit; commit message lists every artefact touched, by ID.

Trivial-fix lane

For typos, formatting, broken links that do not change meaning: skip steps 4, 9, 10. Still bump `last_reviewed`. If unsure whether a fix is trivial, it isn’t; use the full lane.

Status transitions

- MISSING → DRAFT: first content commit.
- DRAFT → STABLE: owner review; complete; no TODO.
- STABLE → STALE: `last_reviewed` older than current milestone freeze, or upstream changed.
- STALE → STABLE: re-review.
- Any → DEPRECATED: superseded.

Anti-patterns (do not)

1. Edit `STABLE` without bumping `last_reviewed`.
2. Paste the same content in two artefacts.
3. Decide in conversation without an ADR.
4. Create an artefact without an inventory row.
5. Quote another artefact’s prose verbatim.
6. Let OQs accumulate without target dates.
7. Mix levels of abstraction in one artefact.

8. Defer propagation (“I’ll update tests later”).
9. Edit a past ADR’s body.
10. Bypass SSOT “just for clarity”.

Session protocol

Warm-up. Read inventory; read protocol TL;DR; read task’s artefacts; spawn supplementary reads. Static L0: keep a project-root bootstrap file — a governed *pointer* to the canonical artefacts, never a restatement — so any agent harness auto-loads the warm-up entry point.

Work. Apply edit protocol; cite by ID; track multi-artefact edits as a unit.

Handoff. Every edit committed; every new artefact has frontmatter and inventory row; decisions in ADRs; questions in OQs; metadata current; inventory’s priority queue reflects new state.

Adoption stack (delegate in this order)

1. L0 warm-up: bootstrap pointer file; agent reads the task’s `depends_on` closure.
2. L1 watchdog: links, inventory, frontmatter schema, staleness auto-degrade, orphans.
3. L2 decision capture: agent drafts ADRs at the moment of decision; human approves.
4. L3 retros and handoffs: drafted from observable state; human edits.
5. L4 graph-native substrate: markdown artefacts as rendered views over a knowledge graph.

Human surface that never delegates: intent, decision approval, scope, voice.